

Article

Tree Growth and Water-Use Efficiency Do Not React in the Short Term to Artificially Increased Nitrogen Deposition

Francesco Giammarchi ^{1,*}, Pietro Panzacchi ^{1,2} , Maurizio Ventura ¹  and Giustino Tonon ¹ 

¹ Faculty of Science and Technology, Free University of Bolzano-Bozen, Piazza Università 5, 39100 Bolzano, Italy; pietro.panzacchi@gmail.com (P.P.); maurizio.ventura@unibz.it (M.V.); giustino.tonon@unibz.it (G.T.)

² Centro di Ricerca per le Aree Interne e gli Appennini (ArIA), Università degli Studi del Molise, Via Francesco De Sanctis 1, 86100 Campobasso, Italy

* Correspondence: francesco.giammarchi@unibz.it

Received: 3 December 2019; Accepted: 27 December 2019; Published: 31 December 2019



Abstract: Increasing atmospheric CO₂ concentration and nitrogen deposition are, among the global change related drivers, those playing a major role on forests carbon sequestration potential, affecting both their productivity and water-use efficiency. Up to now, results are however contrasting, showing that the processes underlying them are far from being fully comprehended. In this study, we adopted an innovative approach to simulate the increase of N deposition in a sessile oak forest in North-Eastern Italy, by fertilizing both from above and below the canopy. We observed the dynamics of basal area increment, intrinsic water-use efficiency and of several leaf functional traits over 4 years, to evaluate how the added nitrogen and the two different fertilization system could affect them. We were not able, however, to detect any shift, besides a common yearly variability related to a prevailing background environmental forcing. To this end, we considered as relevant factors both the short time-span of the observation and the relatively low rate of applied nitrogen. Therefore, we stress the importance of long-term, manipulative experiments to improve the understanding of the C sequestration and mitigation ability of forests in response to increased N deposition.

Keywords: Sessile oak; deciduous forest; carbon sequestration

1. Introduction

Atmospheric carbon dioxide concentration (Ca) increase and nitrogen (N) depositions are considered among the climate change-related drivers that play a major role on forest productivity and carbon (C) sequestration potential [1,2]. However, evidences on the effect of rising CO₂ levels on forest productivity, as well as on the intrinsic water-use efficiency (iWUE) of trees are still contrasting. In fact, although several authors reported a positive relationship between forest growth and iWUE [3–5] especially in drought prone sites an increase in Ca was not followed by a corresponding increase in tree growth, though an increase in iWUE was observed [6,7]. In the last decades, nitrogen oxides (NO_x) emitted during fuel combustion and ammonia volatilization resulting from intensive agriculture have increased atmospheric N deposition, mostly as NO₃[−] and NH₄⁺, especially in the Northern Hemisphere [8]. Similarly to CO₂ increase, N deposition is thought to improve forest productivity and C storage [9], as both temperate and boreal forests are considered N-limited ecosystems [10,11]. However, also concerning the effect of atmospheric N deposition on forest ecosystem, contrasting results are reported in the literature. Some long-term experiments showed that chronic soil N addition could have negative effects on forest growth and soil organic matter mineralization due to the N-saturation process [12]. At the same time, other authors have shown an important stimulating effect of N

deposition on forest biomass accumulation [13] and consequently on the overall ecosystem carbon (C) storage potential [9,14]. Erisman et al. [15], reviewing a number of studies based on experimental data and model simulations, reported C sequestration rates up to 160 kg of C per kg of N added, with most of the results ranging between 35 and 65 kg C/kg N. These conclusions are also supported by the results of long-term manipulation studies, where N has been added to the soil for periods of more than a decade [16]. In these studies, N deposition was found to exert an effect on the water use efficiency at the tree level by increasing either the leaf area [17] or the photosynthetic capacity [18]. Short-term studies showed different responses to increased N inputs, related both to an enhancement of assimilation rate (A) [19,20] and to a reduction of stomatal conductance (g_s) [21]. Overall, the discrepancies of the available research results suggest that more long-term experimental studies are needed for a better understanding of the response of forests to environmental changes and to elucidate the mechanisms behind the sensitivity of forest productivity to N addition. Furthermore, most of the manipulation experiments performed so far have simulated increase in N depositions providing N fertilization on the forest floor, bypassing the tree canopy [22]. However, it has been shown that tree canopy can absorb a large amount of N from atmospheric deposition [23,24]. The passage through the canopy can determine a change in the chemical form of the N input with deposition [24–26]. Results from previous fertilization experiments may therefore have been biased by the absence of the interaction with the canopy, which should be included in experimental designs if the real impact of N depositions on forest ecosystems has to be assessed. Nitrogen deposition is expected to increase in many regions and has been predicted to almost double globally by 2050 [27]. Therefore, to know the sensitivity of forest ecosystem to N deposition is crucial to guide forest managers' choices in times to come. In this study, we present the first results of a long-term experiment, where N was added to the forest both from above and below the canopy. Specifically, the aim was to assess the short-term effects of the differential N fertilization systems on forest growth, iWUE and on several leaf functional traits.

2. Results

The correlation matrix shows a significant relation between iWUE and leaf area index, (LAI), as well as between LAI and canopy N content and relative leaf N concentration (Table 1). The other parameters do not result to be significantly correlated. No significant correlations are evidenced when the dataset is split by single treatment.

Table 1. Correlation matrix for basal area increment (BAI), intrinsic water use efficiency (iWUE), relative leaf nitrogen content (Nleaves), leaf area index (LAI), leaf mass per area (LMA) and canopy nitrogen content (Ncanopy). For each variable in the first row are the Pearson correlation coefficients, whereas in the second row, in italic, are the significance values ($p < 0.05$). Significant correlations are highlighted in bold characters.

	BAI	iWUE	LAI	LMA	Ncanopy	Nleaves
BAI		0.0333 <i>0.8472</i>	0.2139 <i>0.2104</i>	−0.0269 <i>0.8674</i>	0.2658 <i>0.1171</i>	0.1880 <i>0.2722</i>
iWUE	0.0333 <i>0.8472</i>		−0.3310 0.0487	−0.1426 <i>0.4068</i>	−0.2403 <i>0.1580</i>	0.3178 <i>0.0589</i>
LAI	0.2139 <i>0.2104</i>	−0.3310 0.0487		−0.2953 <i>0.0804</i>	0.5895 0.0002	−0.0401 <i>0.8165</i>
LMA	−0.0269 <i>0.8674</i>	−0.1426 <i>0.4068</i>	−0.2953 <i>0.0804</i>		0.3290 <i>0.2060</i>	0.0746 <i>0.6654</i>
Ncanopy	0.2658 <i>0.1171</i>	−0.2403 <i>0.1580</i>	0.5895 0.0002	0.3290 <i>0.2060</i>		0.6114 0.0001
Nleaves	0.1880 <i>0.2722</i>	0.3178 <i>0.0589</i>	−0.0401 <i>0.8165</i>	0.0746 <i>0.6654</i>	0.6114 0.0001	

Regardless of the fertilization method, the basal area increments obtained from girth tape measurements present a clear inter-annual variability with higher values in 2016 and 2018, which are common to all the plots (Figure 1a). According to the general linear model (GLM) repeated measurements, the effect of the N addition was never significant, whereas the time, i.e., the years, resulted to be a highly significant factor (Table 2). Moreover, no significant interaction between year and treatment was detected for this variable. The same pattern emerges when taking into consideration the iWUE, the leaves N content, as well as the total canopy N content (Figure 1b–f), though with different trends among the single variable.

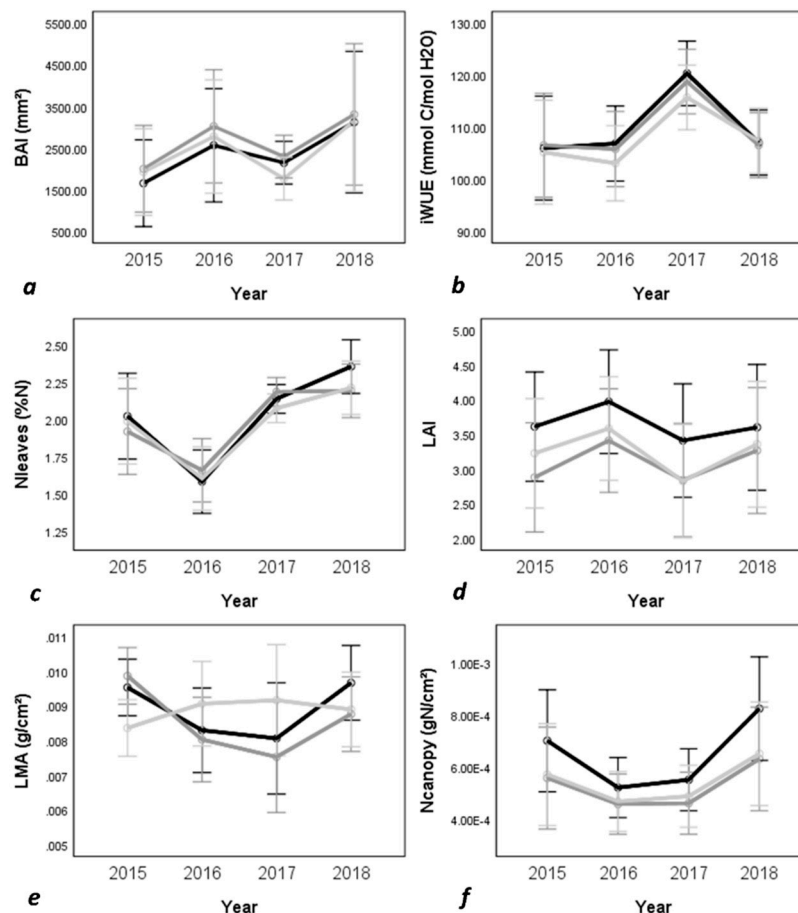


Figure 1. Trend of basal area increment (BAI, **a**), intrinsic water use efficiency (iWUE, **b**), relative leaf nitrogen content (Nleaves, **c**), leaf area index, (LAI, **d**), leaf mass per area (LMA, **e**) and canopy nitrogen content (Ncanopy, **f**), observed in the three different treatments during the 4 years observation period. Data points represent mean values of three plots. SE are shown in vertical bars. Black line = above canopy fertilization; dark grey line = below canopy fertilization; light grey = control).

Table 2. Results for the general linear model (GLM) repeated measures, considering the treatment as between-subject factor and years as within-subject factor. The interaction between the two is also shown. Significant statistics ($p < 0.05$) are highlighted in bold characters. Variables are: basal area increment (BAI), intrinsic water use efficiency (iWUE), relative leaf nitrogen content (Nleaves), leaf area index (LAI), leaf mass per area (LMA) and canopy nitrogen content (Ncanopy).

	Variable	F	<i>p</i>
Treatment	BAI	0.119	0.890
	iWUE	0.228	0.803
	Nleaves	0.180	0.839
	LAI	1.936	0.225
	LMA	2.060	0.208
	Ncanopy	2.060	0.208
Year	BAI	16.985	0.000
	iWUE	21.986	0.000
	Nleaves	62.191	0.000
	LAI	2.867	0.065
	LMA	2.925	0.062
	Ncanopy	11.638	0.000
Year × Treatment	BAI	0.358	0.896
	iWUE	0.310	0.923
	Nleaves	1.097	0.401
	LAI	0.156	0.985
	LMA	2.324	0.077
	Ncanopy	0.406	0.865

The iWUE displayed a particularly high value in 2017. The leaf N concentration is decreasing in 2016 and then constantly increasing in 2017 and 2018, whereas canopy N content decreased in 2016, remained constant in 2017 and increased in 2018. Values of LAI were slightly higher in 2016, while LMA was decreasing in 2016 and 2017 and increasing in 2018 in the treated plots, but remained more or less constant in the control plots throughout the whole period. However, for both LAI and LMA, the effect of year was not significant, as well as that of the treatment and the interaction between time and treatment (Table 2).

3. Discussion

The aim of this study was to evaluate the short-term effects of N addition by using two different N fertilization systems, above and below the canopy, on tree growth, intrinsic water-use efficiency at leaf level and on several leaves functional traits. To our knowledge, very few experimental studies explored the effects of N depositions on forest ecosystems through canopy fertilization [24,28,29]. Even less did it by comparing N fertilizer addition on the ground and above the canopy [22], or by investigating the iWUE changes after N fertilizations through stable isotope analyses [30]. Most probably, the reasons behind this deficiency are the costs, effort required and intrinsic difficulties of such an approach. However, this leads to a crucial lack of knowledge regarding the way that N depositions naturally occur. In fact, the amount of N intercepted by the canopy can represent a significant part of the overall N reaching the ecosystem. For instance, Gaige et al. [24] showed that, in a North-American coniferous forest, the canopy layer retains from 57% to 75% of the NO_3^- and from 73% to 83% of the NH_4^+ . Thus, neglecting this portion of the overall N cycle can lead to relevant biases in the understanding of the forest ecosystem functioning.

Depending on tree species and original nutritional status, nitrogen deposition was found to affect plant transpiration and stomata conductance on leaf, tree and stand scales influencing in turn net carbon assimilation and climate change mitigation potential [31]. However, the existing studies reported often contrasting results and the mechanisms that are behind the response of forests to N depositions

in terms of water balance and C gain are still matter of debate [15,32]. The balance between C gain and water loss by transpiration is well represented by the iWUE, being the ratio of assimilation rate to stomatal conductance. In this context, N deposition was found to increase iWUE in *Quercus velutina* and *Populus × euroamericana* [19,30] to have no effect in *Fagus sylvatica* L. and *Pinus sylvestris* L. [21,33], and to decrease iWUE of phosphorus-limited tropical forests [34]. N directly reaching the canopy is readily available for uptake by the leaves, particularly in the form of NH_4^+ [24] and its assimilation is expected to increase leaf N content, affecting photosynthesis [19] and ultimately the C sequestration potential. Therefore, the magnitude of the interception might have a greater impact on the C cycle and on the potential of forest ecosystems to act as carbon sink [9]. In fact, photosynthetic rate was found to be strictly correlated to leaf N content in many studies and generally associated with an increased rubisco activity [35]. Since it has been observed that the canopy reacts faster to N additions [20], in our study we were expecting to find already in the short-term a divergent response of the N_{AB} treated plots when compared to the N_{BL} plots. However, no clear effect of the treatment was found and only a temporal trend was detected, regardless of the treatment, which likely points to a still prevailing role of the local conditions in determining the variations of the investigated parameters. For instance, we observed a common and consistent trend of iWUE for all the treatments that could be ascribed to local climatic variability, while the influence of the N addition on the carboxylation [35], and hence on the A component of the iWUE, could still be masked by the former. The magnitude of the trees' reaction to N fertilization can be mostly ascribed to the stand species mixture and age, the background deposition as well as the amount and duration of N addition [36]. In this study, the absence of evident effects might be related to the time needed for an ecosystem such as an oak adult high-forest to shift its N cycle in response to added N, applied either above or below canopy. Among the studies that assessed the relationship between N, C and water cycles, Jennings et al. [30] investigated the behavior of both iWUE and tree growth on a 60 years old black oak forest following long-term manipulative fertilizations, and showed that there was a positive effect on both. Similar findings were also obtained by Cornejo-Oviedo et al. [37] on a 20 years old Douglas fir plantation. Here, N fertilization had a positive impact on iWUE, the increase of which promoted tree growth. The results of these studies, however, were gathered respectively after 23 years and after 7 years since the start of the experiment, whereas the time span of our study consists of only 4 years of observations after the beginning of the treatments. On the other hand, Guerrieri et al. [20] found a positive response of the iWUE in a Sitka spruce plantation already within 5 years. Nevertheless, this suggests that both the age of the stand and the ecology of the studied species might play a significant role. Our forest is, in fact, similar to the former case, being all the stands at around 80 years of age, whereas the latter two are young plantations of fast-growing species, in which the elevated N inputs applied (224 kg N/ha in the Douglas fir stand) likely triggered a much faster response. At this point, a second issue must be taken into consideration, i.e., the amount of added N. Though in this study this is not low if compared to the background N deposition levels in the Monticelo site, it could be a further possible explanation of the lack of a short-term response, especially when compared to similar manipulative experiments. Different studies reported that only high levels of N addition would result in higher leaf N content [38–40], with different species and forests being insensitive to low inputs of N deposition [41–43]. In line with the above-mentioned studies, both leaf N content and total canopy N content were unaffected by the external N addition in our experiment. This result helps to explain the absence of physiological response to nitrogen treatments in terms of both WUE and growth rate that we observed in our study. Besides the above-mentioned research of Cornejo-Oviedo et al. [37], other studies focusing on either below canopy fertilization or on both above and below addition normally applied greater quantities of N. For instance, Sheppard et al. [44] applied N at two different rates (48 and 96 Kg N ha⁻¹ yr⁻¹); Jennings et al. [30] applied 50 and 150 kg N ha⁻¹ yr⁻¹ and Zhang et al. [22] 25 and 50 kg N ha⁻¹ yr⁻¹. Only Gaige et al. [24] applied a quantity similar to the one added in this study (18–20 kg N ha⁻¹ yr⁻¹), though for a longer period than in our experiment. This suggests that 4 years is still a short time to allow observing a clear reaction of the studied forest to the N addition, regardless of the fertilization

system. To this end, our results show that the leaves N% was changing in a common way among the treatments almost all the time, with the N_{AB} showing a higher value only in 2018, though not significant. Moreover, not overly elevated N addition rates and a less fast responding forest ecosystem might have easily interacted and, combined, contributed to the observed lack of effects on both tree growth, iWUE and the leaves functional traits.

4. Material and Methods

4.1. Study Site and Experimental Design

The study area is located in Monticolo (46°25'35'' N; 11°17'55'' E), in the Autonomous Province of Bolzano, Italy, at about 550 m above sea level (a.s.l.). The forest stand in which the plots have been established is composed of Sessile oak (*Quercus petraea* L.), up to 97%. Minor species are Scots pine (*Pinus sylvestris* L.), sweet chestnut (*Castanea sativa* Mill.) and several others occurring sporadically, such as lime (*Tilia cordata* Mill.), European hop-hornbeam (*Ostrya carpinifolia* Scop.) and silver birch (*Betula pendula* Roth). The main biometric characteristics of the forest stand at the beginning of the experiment are reported in Table 3.

Table 3. Species mixture, total basal area and stand characteristics of the forest stand at the beginning of the experiment. Stand characteristics referred to sessile oak population only (g_m : average basal area; d_{mg} : diameter at average basal area; h_{mg} : height at average basal area diameter; Hd: dominant height (average height of the tallest 100 trees); V: standing volume).

	N Trees		G (Basal Area)		Stand Characteristics	
	N/ha	%	m ² /ha	%		
<i>Quercus petraea</i>	1125	95.62%	26.969	95.31%	g_m (m ² /ha)	0.02
<i>Tilia cordata</i>	20	1.67%	0.260	0.92%	d_{mg} (cm)	17.50
<i>Castanea sativa</i>	2	0.21%	0.005	0.02%	H_{mg} (m)	13.37
<i>Ostrya carpinifolia</i>	12	1.04%	0.147	0.52%	Hd (m)	16.40
<i>Pinus sylvestris</i>	15	1.25%	0.786	2.78%	V (m ³ /ha)	215.47
<i>Acer spp.</i>	2	0.21%	0.001	0.01%		
Total	1177		28.168			

The forest lies on an acid brown soil originating from porphyritic quartz rock. Average annual temperature at the site is 11.4 °C, whereas average annual precipitation is 800 mm [45]. The experimental design consisted of a set of nine circular plots (12 m radius), three for each different treatment: unfertilized control plot, fertilization by adding $NH_4^+NO_3^-$ directly to the ground (i.e., traditional fertilization, N_{BL}) and fertilization by adding $NH_4^+NO_3^-$ to the canopy by aerial mist spraying (i.e., aerial fertilization, N_{AB} ; Figure 2). The nine plots were arranged in a completely randomized design and were established close to each other, in an area of about 200 m of length, but with at least 10 m of distance between each plot. This was chosen in order to minimize the variation of stand conditions while avoiding issues related to drifting of the fertilizer during applications. The $NH_4^+NO_3^-$ solution (4.3 g N/L) was applied at ground or canopy level (at 15–18 m height, depending on plot maximum tree height) by aerial mist spraying, resulting in a deposition of 20 kg ha⁻¹ yr⁻¹, which was more than three times higher than background atmospheric N deposition [45]. A gasoline powered pump (Officine Carpi S.R.L., Poviglio, Reggio Emilia, Italy) was used to ensure the needed pressure. The pump was then connected, for aerial fertilization, to rotating sprinklers (Rain Bird SNC, Aix-en-Provence, France) mounted on telescopic posts (Fireco S.R.L., Gussago, Brescia, Italy) installed at the center of the N_{ab} plots. These sprinklers provided a spray radius of 12 m, covering the whole plot when operating at a pressure of 2 bars. Tests on the uniformity of the quantity of water provided in the covered area were performed before installation. For N_{BL} plots, the $NH_4^+NO_3^-$ solution was applied manually with a water hose and a spray nozzle, paying attention to uniformly distribute the solution in the treated area. The fertilization was provided in five application dates, monthly from May to September starting from

May 2015. The amount of water provided in each plot with fertilization ($210 \text{ L H}_2\text{O yr}^{-1}$) is equivalent to a precipitation of 0.46 mm for the season, which is negligible if compared to the average annual precipitation of the region.

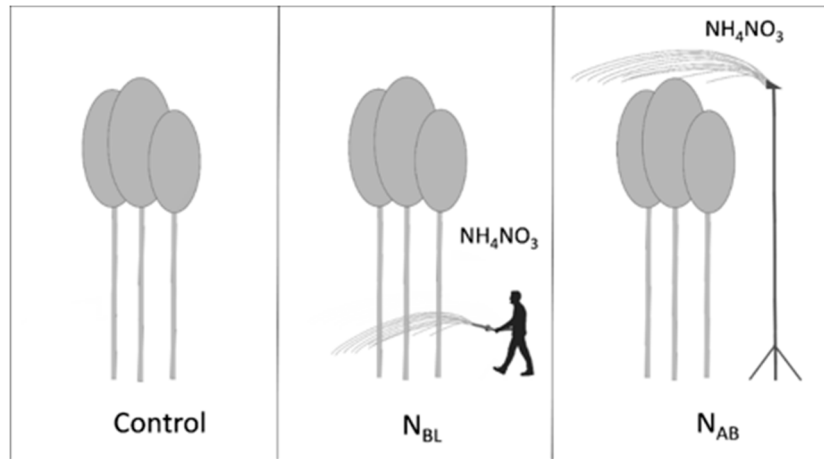


Figure 2. Illustrated scheme of the fertilization system adopted in this study. N_{BL} = below canopy fertilization; N_{AB} = above canopy fertilization; control = no fertilization.

4.2. Tree Growth

At the beginning of the experiment, in 2014, we measured all diameters of each tree within the plots using a tree caliper, with a threshold of 5 cm at breast height (i.e., DBH, 1.30 m). Successively, we placed permanent girth tapes D1 (UMS Gbh, München, DE, Germany) on a subsample of 10 trees per each plot, stratified according to the relative distribution of trees within the DBH classes, in order to obtain measurements as representative as possible of the whole stand growth (see Table S1 for information about number of trees, mean diameter and coefficient of variation for each plot). We then read the girth tapes values every year before the start of the growing season (January or February), from 2015 to 2019. The obtained radial increment values were then transformed in basal area increment (BAI) values, referred both to the plot surface and to the hectare.

4.3. *iWUE* and Leaves Parameters Data Collection

Discrimination against the heavier C stable isotope occurs during the photosynthetic process. Therefore, plant tissues are generally depleted in ^{13}C if compared to the atmospheric CO_2 and the ^{13}C isotopic signature ($\delta^{13}\text{C}$) in leaves reflects changes in the c_i/c_a ratio, which is the ratio between CO_2 concentration in the leaf intercellular spaces (c_i) and in the atmosphere (c_a). These are linked to changes in both A and g_s . This relationship is described in the simplified Farquhar equation as follows [46]:

$$\delta^{13}\text{C}_p = \delta^{13}\text{C}_a - a - (b - a) (c_i/c_a), \quad (1)$$

where $\delta^{13}\text{C}_a$ is the isotopic signature of atmospheric CO_2 , a is the fractionation for $^{13}\text{CO}_2$ during diffusion through air (4.4‰) and b is the fractionation occurring during carboxylation (27‰) by the Rubisco enzyme. From this equation, it is possible to derive c_i as follows:

$$c_i = c_a ((\delta^{13}\text{C}_a - \delta^{13}\text{C}_p - a)/(b - a)). \quad (2)$$

Therefore, given the values of CO_2 atmospheric concentration and those of atmospheric and leaf $\delta^{13}\text{C}$, it is possible to estimate the intrinsic water-use efficiency (*iWUE*; A/g_s) by the following equation:

$$iWUE = A/g_s = (c_a - c_i)/1.6 = (c_a - (c_a (\delta^{13}\text{C}_a - \delta^{13}\text{C}_p - a)/(b - a)))/1.6. \quad (3)$$

Oak leaves were sampled in July of four consecutive years (2015–2018) for stable isotope analysis and both relative N concentration and leaf mass per area (LMA) determination. We randomly sampled 30 fresh leaves in each plot, taken from several trees and only from the part of the canopies directly exposed to light radiation through telescopic pruning shears. Immediately after the sampling, we weighted each single fresh leaf and measured the respective leaf area through a LI-COR 3000C (LI-COR, Lincoln, NE, USA), in order to determine the LMA. Afterwards, the leaves were oven dried overnight at 60 °C, pooled for each plot and milled. The obtained powder was then used for the $\delta^{13}\text{C}$ isotopic analyses. The analysis was done by using an EA elemental analyzer (Flash 2000, Thermo Scientific, Waltham, MA, USA) connected to an isotope ratio mass spectrometer (Delta V Advantage, Thermo Scientific, Waltham, MA, USA) via a continuous flow interface (ConFlo IV, Thermo Scientific, Waltham, MA, USA). Isotope ratios were expressed as permil δ notation ($\delta = (R_{\text{sample}}/R_{\text{standard}}) - 1 \times 1000$) relative to VPDB international standards for ^{13}C . Moreover, $\delta^{13}\text{C}$ values were corrected for the “Suess effect” [47], i.e., the decrease of atmospheric $\delta^{13}\text{C}$ due to emissions of ^{13}C depleted carbon dioxide since the onset of industrialization. Data for correction were retrieved from NOAA mean annual CO_2 data (www.esrl.noaa.gov/gmd/ccgg/trends). iWUE values were calculated using Equation (3). Together with the isotope signature of leaves, also N content was measured.

The total leaf area index (LAI) per plot was determined from litterfall quantification, by means of three litter traps with a collecting area of 1590 cm² each, placed in every plot and surveyed periodically each year from November to January. Eventually, by multiplying the N concentration by LMA and LAI we obtained the canopy total N content (g N/cm²).

4.4. Data Analysis

The results obtained in terms of tree growth (BAI), iWUE and leaves functional traits (leaves N content, LMA, LAI and canopy total N) were related through a correlation analysis (IBM SPSS 25, IBM, Armonk, NY, USA) by pooling all the data together and by splitting the data-set by N addition treatment (N_{AB} , N_{BL} and N_{CTRL}), including all sampled years. To understand at what extent the different treatments affected the changes of the considered features over time, we applied a general linear model (GLM) repeated measures procedure, using IBM SPSS 25 software (IBM, Armonk, NY, USA), including all the considered variables, with the sampling years as within-subject factor and the fertilization treatment as between-subject factors, with a significant level of 0.05.

5. Conclusions

After 4 years of additional N supply, both above and below the canopy, we were not able to detect an effect on tree growth, iWUE and some leaves functional traits. On the other hand, we observed a common pattern for almost all the investigated parameters, hinting to a still prevailing background environmental forcing on the studied stand, rather than that of the simulated N deposition. Among the factors responsible for the lack of response, we considered as relevant both the short time-span of the observation and the relatively low rate of N applied, particularly in relation to the forest species and age. However, given the crucial role of N depositions in affecting the C sequestration potential of temperate forests and their response to increased drought, also the long-term effects of the experiment should be investigated in the future, to further understand the interactions of the above-mentioned factors and the overall forest climate change mitigation potential.

Supplementary Materials: The following are available online at <http://www.mdpi.com/1999-4907/11/1/47/s1>; Table S1: Number of trees, mean diameter and mean diameter coefficient of variation (CV) for each plot at the time of girth tapes placement in 2014.

Author Contributions: P.P., M.V. and G.T. conceived the experiment. F.G., P.P., M.V. and G.T. performed the experiment. F.G., P.P., M.V. and G.T. analyzed the data. F.G. drafted the manuscript, F.G., P.P., M.V. and G.T. reviewed the manuscript. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Free University of Bolzano, through NITROFOR (grant number 141J12000820005) and DECANITRO (grant number I52F15000170005) research projects.

Acknowledgments: We would like to thank Stefano Minerbi of the Bolzano Province Forest Service for assistance in the selection of the study site as well as Joel Towoua and Lorenzo Panizzon for fieldwork assistance.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Hyvönen, R.; Agren, G.I.; Linder, S.; Persson, T.; Cotrufo, M.F.; Ekblad, A.; Freeman, M.; Grelle, A.; Janssens, I.A.; Jarvis, P.G.; et al. The likely impact of elevated [CO₂], nitrogen deposition, increased temperature and management on carbon sequestration in temperate and boreal forest ecosystems: A literature review. *New Phytol.* **2007**, *173*, 463–480. [[CrossRef](#)] [[PubMed](#)]
2. Core Writing Team; Pachauri, R.K.; Meyer, L.A. (Eds.) *Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*; IPCC: Geneva, Switzerland, 2014; p. 151.
3. Gagen, M.; Finsinger, W.; Wagner-Cremer, F.; McCarroll, D.; Loader, N.J.; Robertson, I.; Jalkanen, R.; Young, G.; Kirchhefer, A. Evidence of changing intrinsic water-use efficiency under rising atmospheric CO₂ concentrations in Boreal Fennoscandia from subfossil leaves and tree ring $\delta^{13}\text{C}$ ratios. *Glob. Chang. Biol.* **2011**, *17*, 1064–1072. [[CrossRef](#)]
4. Leonardi, S.; Gentilella, T.; Guerrieri, R.; Ripullone, F.; Magnani, F.; Mencuccini, M.; Noije, T.V.; Borghetti, M. Assessing the effects of nitrogen deposition and climate on carbon isotope discrimination and intrinsic water-use efficiency of angiosperm and conifer trees under rising CO₂ conditions. *Glob. Chang. Biol.* **2012**, *18*, 2925–2944. [[CrossRef](#)] [[PubMed](#)]
5. Giammarchi, F.; Cherubini, P.; Pretzsch, H.; Tonon, G. The increase of atmospheric CO₂ affects growth potential and intrinsic water-use efficiency of Norway spruce forests: Insights from a multi-stable isotope analysis in tree rings of two Alpine chronosequences. *Trees-Struct. Funct.* **2017**, *31*, 503–515. [[CrossRef](#)]
6. Andreu-Hayles, L.; Planells, O.; Gutiérrez, E.; Muntan, E.; Helle, G.; Anchukaitis, K.J.; Schleser, G.H. Long tree-ring chronologies reveal 20th century increases in water-use efficiency but no enhancement of tree growth at five Iberian pine forests. *Glob. Chang. Biol.* **2011**, *17*, 2095–2112. [[CrossRef](#)]
7. Peñuelas, J.; Canadell, J.G.; Ogaya, R. Increased water-use efficiency during the 20th century did not translate into enhanced tree growth. *Glob. Ecol. Biogeogr.* **2011**, *20*, 597–608. [[CrossRef](#)]
8. Galloway, J.N. The global nitrogen cycle: Past, present and future. *Sci. China Ser. C Life Sci. Chin. Acad. Sci.* **2005**, *48*, 669–677.
9. Magnani, F.; Mencuccini, M.; Borghetti, M.; Berbigier, P.; Berninger, F.; Delzon, S.; Grelle, A.; Hari, P.; Jarvis, P.G.; Kolari, P.; et al. The human footprint in the carbon cycle of temperate and boreal forests. *Nature* **2007**, *447*, 848–850. [[CrossRef](#)]
10. Cleveland, C.C.; Townsend, A.R.; Schimel, D.S.; Fisher, H.; Howarth, R.W.; Hedin, L.O.; Perakis, S.S.; Latty, E.F.; Von Fischer, J.C.; Hlseroad, A.; et al. Global patterns of terrestrial biological nitrogen (N₂) fixation in natural ecosystems. *Glob. Biogeochem. Cycles* **1999**, *13*, 623–645. [[CrossRef](#)]
11. LeBauer, D.S.; Treseder, K.K. Nitrogen Limitation of Net Primary Productivity in Terrestrial Ecosystems Is Globally Distributed. *Ecology* **2008**. [[CrossRef](#)]
12. Aber, J.; McDowell, W.; Nadelhoffer, K.; Magill, A.; Berntson, G.; Kamakea, M.; McNulty, S.; Currie, W.; Rustad, L.; Fernandez, I. Nitrogen Saturation in Temperate Forest Ecosystems. *Bioscience* **1998**, *48*, 921–934. [[CrossRef](#)]
13. Townsend, A.R.; Braswell, B.H.; Holland, E.A.; Penner, J.E. Spatial and Temporal Patterns in Terrestrial Carbon Storage Due to Deposition of Fossil Fuel Nitrogen. *Ecol. Appl.* **1996**, *6*, 806–814. [[CrossRef](#)]
14. Sutton, M.A.; Simpson, D.; Levy, P.E.; Smith, R.I.; Reis, S.; van Oijen, M.; de Vries, W. Uncertainties in the relationship between atmospheric nitrogen deposition and forest carbon sequestration. *Glob. Chang. Biol.* **2008**, *14*, 2057–2063. [[CrossRef](#)]
15. Erisman, J.W.; Van Grinsven, H.; Grizzetti, B.; Bouraoui, F.; Powlson, D.; Sutton, M.A.; Bleeker, A.; Reis, S. The European nitrogen problem in a global perspective Executive summary Major uncertainties/challenges. In *The European Nitrogen Assessment*; Sutton, M.A., Britton, C., Erisman, J.W., Billen, G., Bleeker, A., Greenfelt, P., van Grinsven, H., Grizzetti, B., Eds.; Cambridge University Press: Cambridge, UK, 2011; pp. 9–31.
16. Hyvönen, R.; Persson, T.; Andersson, S.; Olsson, B.; Ågren, G.I.; Linder, S. Impact of long-term nitrogen addition on carbon stocks in trees and soils in northern Europe. *Biogeochemistry* **2008**, *89*, 121–137. [[CrossRef](#)]

17. Ewers, B.E.; Oren, R.; Sperry, J.S. Influence of nutrient versus water supply on hydraulic architecture and water balance in *Pinus taeda*. *Plant Cell Environ.* **2000**, *23*, 1055–1066. [\[CrossRef\]](#)
18. Lambers, H.; Chapin, F.S., III; Pons, T. *Plant Physiological Ecology*; Springer: Berlin/Heidelberg, Germany, 2008.
19. Ripullone, F.; Lauteri, M.; Grassi, G.; Amato, M.; Borghetti, M. Variation in nitrogen supply changes water-use efficiency of *Pseudotsuga menziesii* and *Populus x euroamericana*; a comparison of three approaches to determine water-use efficiency. *Tree Physiol.* **2004**, *24*, 671–679. [\[CrossRef\]](#)
20. Guerrieri, R.; Mencuccini, M.; Sheppard, L.J.; Saurer, M.; Perks, M.P.; Levy, P.; Sutton, M.A.; Borghetti, M.; Grace, J. The legacy of enhanced N and S deposition as revealed by the combined analysis of $\delta^{13}\text{C}$, $\delta^{18}\text{O}$ and $\delta^{15}\text{N}$ in tree rings. *Glob. Chang. Biol.* **2011**, *17*, 1946–1962. [\[CrossRef\]](#)
21. Betson, N.R.; Johannisson, C.; Löfvenius, M.O.; Grip, H.; Granström, A.; Högborg, P. Variation in the $\delta^{13}\text{C}$ of foliage of *Pinus sylvestris* L. in relation to climate and additions of nitrogen: Analysis of a 32-year chronology. *Glob. Chang. Biol.* **2007**, *13*, 2317–2328. [\[CrossRef\]](#)
22. Zhang, W.; Yan, J.; Zhang, W.; Shen, W.; Zhu, S.; Cai, X.; Liu, Z.; Wang, F.; Rao, X.; Mo, J.; et al. CAN Canopy Addition of Nitrogen Better Illustrate the Effect of Atmospheric Nitrogen Deposition on Forest Ecosystem? *Sci. Rep.* **2015**, *5*, 1–12. [\[CrossRef\]](#)
23. Sievering, H.; Tomaszewski, T.; Torizzo, J. Canopy uptake of atmospheric N deposition at a conifer forest: Part I—canopy N budget, photosynthetic efficiency and net ecosystem exchange. *Tellus Ser. B Chem. Phys. Meteorol.* **2007**, *59*, 483–492. [\[CrossRef\]](#)
24. Gaige, E.; Dail, D.B.; Hollinger, D.Y.; Davidson, E.A.; Fernandez, I.J.; Sievering, H.; White, A.; Halteman, W. Changes in canopy processes following whole-forest canopy nitrogen fertilization of a mature spruce-hemlock forest. *Ecosystems* **2007**, *10*, 1133–1147. [\[CrossRef\]](#)
25. Guerrieri, R.; Vanguelova, E.I.; Michalski, G.; Heaton, T.H.E.; Mencuccini, M. Isotopic evidence for the occurrence of biological nitrification and nitrogen deposition processing in forest canopies. *Glob. Chang. Biol.* **2015**, *21*, 4613–4626. [\[CrossRef\]](#) [\[PubMed\]](#)
26. Houle, D.; Marty, C.; Duchesne, L. Response of canopy nitrogen uptake to a rapid decrease in bulk nitrate deposition in two eastern Canadian boreal forests. *Oecologia* **2015**, *177*, 29–37. [\[CrossRef\]](#) [\[PubMed\]](#)
27. Galloway, J.N.; Dentener, F.J.; Capone, D.G.; Boyer, E.W.; Howarth, R.W.; Seitzinger, S.P.; Asner, G.P.; Cleveland, C.C.; Green, P.A.; Holland, E.A.; et al. Nitrogen Cycles: Past, Present, and Future. *Biogeochemistry* **2004**, *70*, 153–226. [\[CrossRef\]](#)
28. Cape, J.N.; Sheppard, L.J.; Crossley, A.; Van Dijk, N.; Tang, Y.S. Experimental field estimation of organic nitrogen formation in tree canopies. *Environ. Pollut.* **2010**, *158*, 2926–2933. [\[CrossRef\]](#) [\[PubMed\]](#)
29. Gilliam, F.S.; Adams, M.B. Effects of Nitrogen on Temporal and Spatial Patterns of Nitrate in Streams and Soil Solution of a Central Hardwood Forest. *ISRN Ecol.* **2011**, *2011*, 1–9. [\[CrossRef\]](#)
30. Jennings, K.A.; Guerrieri, R.; Vadeboncoeur, M.A.; Asbjornsen, H. Response of *Quercus velutina* growth and water use efficiency to climate variability and nitrogen fertilization in a temperate deciduous forest in the northeastern USA. *Tree Physiol.* **2016**, *36*, 428–443. [\[CrossRef\]](#)
31. Fleischer, K.; Rebel, K.T.; Van Der Molen, M.K.; Erisman, J.W.; Wassen, M.J.; Van Loon, E.E.; Montagnani, L.; Gough, C.M.; Herbst, M.; Janssens, I.A.; et al. The contribution of nitrogen deposition to the photosynthetic capacity of forests. *Global Biogeochem. Cycles* **2013**, *27*, 187–199. [\[CrossRef\]](#)
32. Bobbink, A.R.; Hicks, K.; Galloway, J.; Spranger, T.; Alkemade, R.; Ashmore, M.; Cinderby, S.; Davidson, E.; Dentener, F.; Emmett, B.; et al. Global assessment of nitrogen deposition effects on terrestrial plant diversity: A synthesis. *Ecol. Appl.* **2010**, *20*, 30–59. [\[CrossRef\]](#)
33. Elhani, S.; Guehl, J.M.; Nys, C.; Picard, J.F.; Dupouey, J.L. Impact of fertilization on tree-ring $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ in beech stands: A retrospective analysis. *Tree Physiol.* **2005**, *25*, 1437–1446. [\[CrossRef\]](#)
34. Huang, Z.; Liu, B.; Davis, M.; Sardans, J.; Peñuelas, J.; Billings, S. Long-term nitrogen deposition linked to reduced water use efficiency in forests with low phosphorus availability. *New Phytol.* **2016**, *210*, 431–442. [\[CrossRef\]](#) [\[PubMed\]](#)
35. Kattge, J.; Knorr, W.; Raddatz, T.; Wirth, C. Quantifying photosynthetic capacity and its relationship to leaf nitrogen content for global-scale terrestrial biosphere models. *Glob. Chang. Biol.* **2009**, *15*, 976–991. [\[CrossRef\]](#)
36. Pardo, L.H.; Robin-Abbott, M.J.; Driscoll, C.T. Assessment of nitrogen deposition effects and empirical critical loads of nitrogen for ecoregions of the United States. *Gen. Tech. Rep. NRS* **2011**, *80*, 291.

37. Cornejo-Oviedo, E.H.; Voelker, S.L.; Mainwaring, D.B.; Maguire, D.A.; Meinzer, F.C.; Brooks, J.R. Basal area growth, carbon isotope discrimination, and intrinsic water use efficiency after fertilization of Douglas-fir in the Oregon Coast Range. *For. Ecol. Manag.* **2017**, *389*, 285–295. [[CrossRef](#)]
38. Magill, A.H.; Aber, J.D.; Berntson, G.M.; McDowell, W.H.; Nadelhoffer, K.J.; Melillo, J.M.; Steudler, P. Long-term nitrogen additions and nitrogen saturation in two temperate forests. *Ecosystems* **2000**, *3*, 238–253. [[CrossRef](#)]
39. Pitcairn, C.E.R.; Leith, I.; Fowler, D.; Hargreaves, K.; Moghaddam, M.; Kennedy, V.; Granat, L. Foliar Nitrogen as an Indicator of Nitrogen Deposition and Critical Loads Exceedance on a European Scale. *Water Air Soil Pollut.* **2001**, *130*, 1037–1042. [[CrossRef](#)]
40. Talhelm, A.F.; Pregitzer, K.S.; Burton, A.J. No evidence that chronic nitrogen additions increase photosynthesis in mature sugar maple forests. *Ecol. Appl.* **2011**, *21*, 2413–2424. [[CrossRef](#)]
41. Aber, J.D.; Goodale, C.L.; Ollinger, S.V.; Smith, M.-L.; Magill, A.H.; Martin, M.E.; Hallett, R.A.; Stoddard, J.L. Is Nitrogen Deposition Altering the Nitrogen Status of Northeastern Forests? *Bioscience* **2003**, *53*, 375. [[CrossRef](#)]
42. Bauer, G.A.; Bazzaz, F.A.; Minocha, R.; Long, S.; Magill, A.; Aber, J.; Berntson, G.M. Effects of chronic N additions on tissue chemistry, photosynthetic capacity, and carbon sequestration potential of a red pine (*Pinus resinosa* Ait.) stand in the NE United States. *For. Ecol. Manag.* **2004**, *196*, 173–186. [[CrossRef](#)]
43. Hu, Y.; Zhao, P.; Zhu, L.; Zhao, X.; Ni, G.; Ouyang, L.; Schäfer, K.V.R.; Shen, W. Responses of sap flux and intrinsic water use efficiency to canopy and understory nitrogen addition in a temperate broadleaved deciduous forest. *Sci. Total Environ.* **2019**, *648*, 325–336. [[CrossRef](#)]
44. Sheppard, L.J.; Coward, P.; Skiba, U.; Harvey, F.J.; Crossley, A.; Ingleby, K. Effects of five years of frequent N additions, with or without acidity, on the growth and below-ground dynamics of a young Sitka spruce stand growing on an acid peat: Implications for sustainability. *Hydrol. Earth Syst. Sci.* **2010**, *8*, 377–391. [[CrossRef](#)]
45. Marchetti, F.; Tait, D.; Ambrosi, P.; Minerbi, S.; Agrario, I.; Michele, S.; Michele, S.; Tn, A. Atmospheric deposition at four forestry sites in the Alpine Region of Trentino-South Tyrol, Italy. *J. Limnol.* **2002**, *61*, 148–157. [[CrossRef](#)]
46. Farquhar, G. Carbon Isotope Discrimination And Photosynthesis. *Annu. Rev. Plant Physiol. Plant Mol. Biol.* **1989**, *40*, 503–537. [[CrossRef](#)]
47. Francey, R.J.; Allison, C.E.; Etheridge, D.M.; Trudinger, C.M.; Enting, I.G.; Leuenberger, M.; Langenfelds, R.L.; Michel, E.; Steele, L.P. A 1000-year high precision record of delta13C in atmospheric CO2. *Tellus B* **1999**, *51*, 170–193. [[CrossRef](#)]



© 2019 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<http://creativecommons.org/licenses/by/4.0/>).